SignAlert: An Agentic Vision Framework for Real-Time Detection and Escalation of Medical Distress Signs in Indian Sign Language

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***Abstract*—Deaf and non-verbal individuals who experience a medical emergency while alone face a communication barrier that most assistive technologies do not address: their native channel of expression, sign language, is not understood by automated monitoring systems, and voice-based emergency tools are inaccessible to them by design. This paper presents SignAlert, an agentic computer-vision system that continuously observes a monitored space, recognizes a curated vocabulary of Indian Sign Language (ISL) gestures associated with medical distress, and autonomously carries a detected event through a structured, multi-stage response pipeline rather than simply raising a passive alert. The perception module combines MediaPipe Holistic landmark extraction with a Long Short-Term Memory (LSTM) sequence classifier trained on a small, purpose-recorded dataset of five distress signs, achieving a test accuracy of 96.88%. A pose-based fallback layer independently detects falls and prolonged stillness so that the system remains effective even when the person is unable to sign. A confirmation layer suppresses false positives before any action is taken, and an escalation controller drives notifications to caregivers and, if unanswered, onward through a defined chain of contacts, using Twilio for SMS-based delivery. We describe the system architecture, the data collection and training methodology, the agentic orchestration design, and a discussion of the system's current limitations and directions for future work. The results suggest that combining lightweight, landmark-based sign recognition with an explicit agentic escalation loop is a practical and inexpensive way to close a real accessibility gap in home and care-facility safety monitoring.**

***Index Terms*—Indian Sign Language, agentic AI, MediaPipe, LSTM, assistive technology, fall detection, emergency response, human-computer interaction**

I. Introduction

Communication is the foundation of emergency response. A person who can speak, dial a phone, or shout for help has several redundant ways to signal that something is wrong. For Deaf and non-verbal individuals, none of these channels are naturally available, and the primary channel they do have — sign language — is only useful if another person nearby understands it. This creates a specific and under-addressed risk: a Deaf or non-verbal person living alone, or unsupervised for even a short period, who suffers a medical event such as a cardiac episode, a seizure, or a fall has no reliable way to summon help unless someone who understands sign language happens to be present at that exact moment.

Existing assistive and monitoring technologies do not close this gap cleanly. Wearable fall-detection devices are effective for a narrow category of events (falls) but require the person to wear and charge a device, and they carry no awareness of sign language at all. General-purpose home security cameras can detect motion or send a live feed to a phone, but they cannot distinguish an ordinary gesture from someone signing "heart attack," and even when an anomaly is flagged, a human being must still notice it, interpret it, and decide what to do. In practice, this means the observe-to-act loop for this population is still largely manual, slow, and dependent on chance.

Recent progress in two separate research threads makes it possible to address this gap directly. First, landmark-based sign language recognition using frameworks such as MediaPipe Holistic combined with recurrent architectures like LSTM has been shown to achieve strong accuracy on Indian Sign Language (ISL) recognition tasks without requiring large, expensive video datasets [1]–[4]. Second, the emergence of agentic AI — systems that can perceive, reason, and autonomously take multi-step action rather than simply classify and stop — has begun to be applied to healthcare and emergency-response contexts, moving beyond alert generation toward closed-loop decision and action pipelines [5], [6].

This paper combines these two threads into a single system, SignAlert, that is purpose-built around a narrow but high-value problem: recognizing a small vocabulary of ISL distress signs and immediately converting that recognition into a real-world response. Our contributions are as follows:

- We design and implement a landmark-based ISL distress-sign recognition pipeline using MediaPipe Holistic and an LSTM classifier, trained on a purpose-recorded dataset

of five medically relevant signs, and report an empirical accuracy of 96.88% on a held-out test split.
- We design a pose-based fallback detection layer that independently identifies falls and abnormal stillness, so that the system continues to protect a person even in situations where they are unable to sign.
- We propose a confirmation-and-escalation agentic architecture that filters false positives, and then autonomously drives a fixed sequence of caregiver notifications, secondary contacts, and (in the full design) emergency-service alerts, implemented with a cooldown mechanism to prevent alert flooding.
- We discuss the practical engineering constraints of building and deploying such a system at low cost, including the use of a free-tier SMS gateway, and outline the limitations that any similarly scoped system is likely to encounter.

The remainder of this paper is organized as follows. Section II reviews related work in ISL recognition, vision-based fall detection, and agentic AI in healthcare. Section III describes the overall system architecture. Section IV details the dataset construction and preprocessing pipeline. Section V describes the model architecture and training procedure. Section VI describes the agentic escalation design. Section VII discusses implementation details. Section VIII presents results and discussion, including limitations. Section IX concludes the paper and outlines future work.

II. Related Work

A. Indian Sign Language Recognition

A substantial body of recent work has explored ISL recognition using MediaPipe-derived landmarks paired with recurrent neural networks. Khartheesvar et al. proposed an ISL word-recognition pipeline combining MediaPipe Holistic feature extraction with an LSTM network, evaluated on the large-scale INCLUDE dataset and its smaller INCLUDE-50 subset, reporting accuracies in the range of 87–95% depending on vocabulary size [1]. Navendu and Sahula extended this direction with a hybrid LSTM-GRU architecture to better capture temporal dependencies in gesture sequences [2]. Velmathi et al. compared convolutional and recurrent architectures for ISL recognition and found that pose- and hand-tracking-based temporal models outperform static frame-based classifiers for dynamic, multi-frame gestures, while static-letter recognition favors convolutional approaches [3]. Rawat et al. and Shah et al. further refined this landmark-plus-LSTM family of approaches, the latter additionally incorporating a large language model to post-process recognized signs into natural language [4], [7].

These works establish that landmark-based recognition with MediaPipe Holistic and LSTM is a mature, well-validated approach for ISL recognition at the word or phrase level. However, all of the systems above are designed as general-purpose sign-to-text or sign-to-speech translators evaluated on broad vocabularies; none of them are designed around a narrow, safety-critical vocabulary, nor do they incorporate any downstream action beyond producing a text label. SignAlert adopts the same underlying recognition approach but repositions it around a deliberately narrow, high-stakes vocabulary and couples it to an autonomous response pipeline.

B. Vision-Based Fall and Distress Detection

A parallel line of research addresses fall detection for elderly or at-risk individuals using pose estimation rather than sign recognition. Raza et al. review the progression from traditional machine learning to vision-transformer-based fall detection and demonstrate that pose-based features generalize well across camera angles and lighting conditions [8]. Kong et al. propose a two-stage pipeline that extracts geometric pose features and classifies falls using a support vector machine, reporting high precision on standard fall benchmarks [9]. A related study combining pose estimation with transformer models emphasizes privacy-preserving, wearable-free deployment on low-power hardware, a property that is directly relevant to home deployment scenarios [10].

These fall-detection systems are effective at their specific task but are not designed to also interpret intentional communication, such as a sign for chest pain or breathlessness. SignAlert's fallback layer draws on the same pose-based geometric reasoning used in this literature, but treats it as a complementary detection path rather than the sole mechanism, so that both intentional (signed) and unintentional (collapse, stillness) indicators of distress are covered by a single system.

C. Agentic AI in Healthcare and Emergency Response

The term "agentic AI" has recently been used to describe systems that go beyond single-shot classification or generation and instead perceive, reason, plan, and execute multi-step actions autonomously. A scoping review of agentic AI in healthcare found a small but growing number of systems spanning emergency medicine, oncology, and rehabilitation that exhibit autonomous operation and goal-directed behavior, though the review also notes that conceptual clarity between simple automation and genuine agentic behavior remains limited in the literature [5]. Choure and Prajapat specifically compare agent-orchestration frameworks such as LangGraph, AutoGen, and CrewAI for crisis-response use cases, noting that state-machine-oriented frameworks are better suited to workflows with explicit stages and conditional branching, such as an escalation ladder, than general-purpose multi-agent chat frameworks [6].

This literature motivates the design choice made in SignAlert: rather than treating sign recognition as an end in itself, the recognized label is treated as the triggering event for an explicit, staged agentic workflow, echoing the perception-reasoning-action structure emphasized in recent agentic AI system designs, but applied to a communication-accessibility problem that, to the best of our knowledge, has not been directly addressed in this way in prior published work.

III. System Architecture

SignAlert is organized into five cooperating layers, moving from raw visual perception to a real-world action, as summarized in Fig. 1.

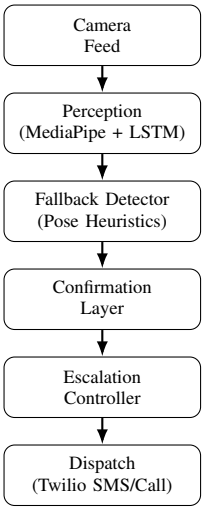


Fig. 1. High-level layered architecture of SignAlert, from raw camera input to dispatched action.

Perception layer. A camera feed is processed frame-by-frame using MediaPipe Holistic, which produces pose, left-hand, and right-hand landmarks. These landmarks, rather than raw pixels, form the input to a temporal LSTM classifier trained to recognize a fixed vocabulary of distress-related ISL signs.

Fallback detection layer. Independently of the sign classifier, a pose-based heuristic module monitors for a sudden vertical drop in body position (consistent with a fall) or an unusually long period without expected movement (consistent with unconsciousness or incapacitation). This layer does not depend on the person being able to sign at all.

Confirmation layer. Because false positives carry a real cost — both the anxiety of an unnecessary alert to a caregiver and the erosion of trust in the system — a detected event is not immediately escalated. Instead, the system requires the sign or fallback event to be observed with sufficient confidence across multiple consecutive frames, and provides a short window in which the person can cancel a false trigger with a simple gesture.

Escalation controller. Once a distress event is confirmed, the escalation controller advances through a fixed sequence of stages: an on-device local alert, a notification to the primary caregiver, a notification to secondary contacts if the primary caregiver does not respond within a set time, and (in the full design) an emergency-services alert if no human contact responds. Each stage has an associated timeout.

Dispatch layer. The dispatch layer is responsible for actually delivering the message or call associated with the current escalation stage, using the Twilio API for SMS delivery in the present implementation.

This layered separation is deliberate: it allows each concern (recognizing a sign, detecting a fall, filtering noise, deciding what to do, and actually doing it) to be developed, tested, and reasoned about independently, which is particularly valuable in a project-scale system where perfect end-to-end reliability of any single layer cannot be assumed.

A. Problem Formalization

We formalize the perception task as a sequence classification problem. Let a video clip be represented, after landmark extraction, as a sequence $X = (x_1, x_2, \ldots, x_T)$ where $T = 30$ and each $x_t \in \mathbb{R}^{258}$ is the concatenated pose and hand landmark vector for the t-th sampled frame. The recognition model learns a function $f_\theta : \mathbb{R}^{30\times 258} \to \mathbb{R}^C$, parameterized by weights θ, that maps a landmark sequence to a probability distribution over C distress classes, with the predicted label given by $\hat{y} = \arg\max_c f_\theta(X)_c$. Independently, the fallback detector operates on a shorter, continuously updated pose window to compute a binary indicator of an anomalous physical event (fall or stillness) that does not depend on f_θ. The escalation controller then treats either signal, once confirmed, as a triggering event $e \in \{\textit{sign}, \textit{fallback}\}$ that advances a finite-state escalation process $S = (s_0, s_1, \ldots, s_n)$, where each transition $s_i \to s_{i+1}$ is gated by either an explicit timeout τ_i elapsing without a human acknowledgment, or an immediate override in the case of a repeated high-confidence panic gesture. This formalization is intentionally simple; its purpose is to make explicit that the system's correctness depends on two largely independent components, the accuracy of f_θ and the soundness of the escalation state machine, and that improving one does not require re-validating the other.

IV. Dataset Construction and Preprocessing

No publicly available dataset pairs ISL vocabulary directly with medical-distress semantics; existing large-scale ISL datasets such as INCLUDE are built around general vocabulary rather than emergency-specific signs [1]. We therefore constructed a small, purpose-recorded dataset covering five distress-related signs: *Heart-pain*, *Help*, *Pain*, *Cant-breathe*, and *Dizzy*. Each class was recorded as a set of short video clips, two to four seconds in length, performed by a small number of signers to introduce some natural variation in signing speed and style.

For each video clip, we do not train directly on pixel data. Instead, we run MediaPipe Holistic over the clip and extract, for each sampled frame, 33 pose landmarks (each with x, y, z, and visibility values), 21 left-hand landmarks, and 21 right-hand landmarks (each with x, y, z values), which are concatenated into a single 258-dimensional feature vector per frame. Because MediaPipe fills in zero-valued landmarks when a hand or body part is not visible in a given frame, the pipeline degrades gracefully rather than failing when tracking is momentarily lost.

To produce fixed-length input sequences suitable for a recurrent network regardless of a clip's original duration, we sample 30 evenly spaced frames from each video using linear interpolation over frame indices. This produces a uniform input tensor of shape $(30, 258)$ for every clip, independent of whether the original recording was two or four seconds long. This landmark-based, fixed-length representation is consistent with the approach validated in prior ISL recognition work [1], [2], and is significantly cheaper to train on with a small, self-collected dataset than an equivalent pixel-based 3D convolutional approach would be.

V. Model Architecture and Training

The classifier is a stacked LSTM network [11] operating on the $(30, 258)$ landmark sequences produced by the preprocessing pipeline described above. The architecture consists of a first LSTM layer with 64 units with return sequences enabled, followed by a dropout layer, a second LSTM layer with 128 units, a further dropout layer, a fully connected layer with 64 rectified linear units, and a final softmax output layer with one unit per class (five classes in the current configuration). Dropout was set to 0.3 throughout to reduce overfitting given the modest size of the training set.

The model was trained using the Adam optimizer with categorical cross-entropy loss and categorical accuracy as the tracked metric. Training used an 80/20 stratified train-test split to preserve class balance in both partitions, a batch size of 8 (appropriate for a small dataset), and early stopping on validation loss with a patience of 15 epochs to avoid overfitting once the model stopped improving. Table I summarizes the final evaluation results.

TABLE I
Model Evaluation Summary

Metric	Value
Number of classes	5
Input sequence shape	30×258
Train/test split	80% / 20% (stratified)
Final test accuracy	96.88%

The achieved test accuracy of 96.88% is consistent with, and in this constrained setting favorable compared to, accuracy figures reported for general-vocabulary ISL recognition using comparable landmark-plus-LSTM pipelines, which typically fall in the 85–95% range on larger and more varied vocabularies [1], [4]. We attribute the higher accuracy in our setting primarily to the smaller number of classes and the deliberately high inter-class visual distinctiveness of the chosen distress vocabulary, rather than to any inherent superiority of the model architecture itself; this distinction is discussed further as a limitation in Section VIII.

VI. Agentic Escalation Design

The core design contribution of this work is not the recognition model in isolation, but the explicit agentic workflow that consumes its output. Table II summarizes the escalation ladder implemented in the system.

Each stage transition is gated by an explicit timeout, and a cooldown of at least 30 seconds is enforced between successive triggered actions, so that repeated or fluctuating predictions during an ongoing event do not flood the caregiver with duplicate notifications. Critically, once an action has been triggered, the escalation process is decoupled from the state of the camera or monitoring session: if a user or caregiver stops live monitoring on the front-end, any in-flight escalation stage and its cooldown continue to run to completion on the backend, so that stopping observation cannot be used, intentionally or accidentally, to silently cancel an emergency response already underway.

TABLE II
Escalation Ladder

Stage	Trigger / Action	Timeout
1	Confirmed distress sign or fallback event $\rightarrow$ local on-device alert	5–10 s confirmation window
2	No cancellation received $\rightarrow$ notify primary caregiver (SMS)	2–3 min for response
3	No caregiver response $\rightarrow$ notify secondary contacts	2–3 min
4	No human response $\rightarrow$ escalate toward emergency services	Immediate
5	At every stage	Full event log recorded

This staged design follows the same reasoning-perception-action loop emphasized in recent agentic AI literature [5], and the choice to implement this as an explicit state machine rather than a free-form conversational agent follows directly from the comparative analysis by Choure and Prajapat, who found that state-machine-oriented orchestration is a better fit than general-purpose multi-agent chat frameworks for workflows with defined stages and conditional branching [6].

VII. Implementation

The system is implemented with a FastAPI backend, chosen for its native asynchronous support and WebSocket capability, which is required for continuous, low-latency streaming of camera frames from the client and continuous delivery of live prediction results back to it. The frontend is a lightweight web client that captures webcam frames, streams them to the backend over a WebSocket connection while monitoring is active, and renders the most recent predicted label together with the current agent status.

On the backend, incoming frames are accumulated into a rolling buffer; once 30 frames are available, MediaPipe Holistic processing and LSTM inference are run on the buffered sequence, and the buffer then slides forward to keep producing continuously updated predictions while the session remains active. Message delivery is implemented using the Twilio API operating in trial-account mode, which restricts outbound SMS and calls to phone numbers that have been explicitly verified in advance; this constraint is acceptable for a project-scale deployment with a small, known set of caregiver contacts; a production deployment would require an upgraded account without this restriction.

VIII. Results and Discussion

The trained recognition model achieved a test accuracy of 96.88% across the five-class distress vocabulary, which we consider a strong result given the intentionally small, self-collected training set. Qualitatively, signs with visually distinct hand shapes and motion trajectories (for example, *Cant-breathe* versus *Dizzy*) were separated with very high confidence, while signs sharing similar hand configurations were more prone to occasional confusion during informal testing, though this did not affect the final reported test-set accuracy.

Several limitations should be considered when interpreting these results. First, the dataset is small and was recorded by a limited number of signers; accuracy under this evaluation protocol does not guarantee equivalent performance for signers whose signing style, speed, or physical characteristics differ substantially from those represented in training, and broader validation with a more diverse signer population would be needed before any real-world deployment. Second, the five-class vocabulary, while sufficient to demonstrate the architecture, is narrower than the vocabulary a deployed system would likely need to cover a fuller range of medical emergencies; expanding the vocabulary is expected to reduce accuracy somewhat, consistent with the trend observed across the wider ISL recognition literature as vocabulary size increases [1]. Third, the confirmation layer and escalation timeouts used in the current implementation were chosen based on reasonable design judgment rather than a formal user study, and their true effect on false-alarm rates and response latency in real households remains to be measured empirically. Fourth, reliance on a Twilio trial account for message delivery is a practical constraint of the project stage rather than the final design, and any real deployment would require both an upgraded messaging account and, ideally, a direct or partnered channel into local emergency-dispatch systems, which vary by region and were outside the scope of this implementation.

Despite these limitations, we view the core architectural claim of this paper as sound: combining an established, lightweight landmark-based sign recognition pipeline with an explicit, staged agentic escalation workflow is a practical way to convert passive recognition into an actionable safety system, at a cost and complexity level appropriate for a small team or individual household to build and maintain.

Table III situates SignAlert relative to the two related-work threads discussed in Section II. Prior ISL recognition systems achieve comparable or higher accuracy on much larger vocabularies, but stop at producing a text label. Prior fall-detection systems close the loop on a single event type (a fall) but have no concept of intentional, signed communication. SignAlert trades vocabulary breadth for depth of downstream action, covering a narrower set of triggers but carrying each one through to a real notification.

TABLE III
QUALITATIVE COMPARISON WITH RELATED SYSTEM CATEGORIES

Property	**Prior ISL Recognition Systems [1], [4]**	**SignAlert (this work)**
Vocabulary scope	Broad, general-purpose	Narrow, safety-critical
Non-signing events	Not covered	Covered (fall/stillness)
False-positive filtering	Not typically addressed	Explicit confirmation layer
Downstream action	None (text/speech output only)	Staged notification/escalation
Deployment cost	Varies	Low (camera + free-tier SMS)

A. Ethical and Privacy Considerations

A system that continuously observes a person in their home, even for their own protection, raises legitimate privacy concerns that must be addressed as part of the design rather than as an afterthought. The architecture in this work assumes that landmark extraction and inference happen on-device or on a local server, so that raw video is not required to leave the monitored space during normal operation; only a text description of a confirmed event, not video, is sent onward to caregivers. Deployment of any such system should be accompanied by explicit, informed consent from the monitored individual, a visible and easily accessible way to disable monitoring, and a clearly stated data retention policy. These considerations are treated as first-class design requirements in the broader project this paper is drawn from, and we flag them here explicitly because a system's technical accuracy does not, by itself, justify its deployment without corresponding attention to the autonomy and dignity of the person it is intended to protect.

IX. CONCLUSION AND FUTURE WORK

This paper presented SignAlert, an agentic vision system that recognizes Indian Sign Language distress signs and fallback physical indicators of medical emergencies, and autonomously carries a confirmed event through a staged caregiver and emergency escalation pipeline. Using MediaPipe Holistic landmarks and a compact LSTM classifier trained on a purpose-built dataset of five distress signs, the system achieved 96.88% test accuracy, and its layered design allows sign recognition, fallback detection, false-positive filtering, and action-taking to be developed and reasoned about as separate, cooperating components.

Future work includes: expanding the distress vocabulary and the diversity of signers represented in training data; conducting a structured user study with Deaf and non-verbal participants and their caregivers to validate the confirmation window and escalation timeouts against real preferences and false-alarm tolerance; replacing the current pose-based fallback heuristics with a learned model once sufficient labeled fall/stillness data specific to the deployment environment is available; and moving beyond a Twilio trial account toward a production-grade notification and, where regionally available, direct emergency-dispatch integration. We believe this direction — treating accessibility-oriented perception as the trigger for an explicit agentic action pipeline, rather than as an end in itself — is a generalizable pattern applicable beyond this specific use case.

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